

Cell3 to 29-Lock Native Witness Framework

Educational successor to the galaxy-sourced cell0, authoritative cell3, and Mass-Bridge derivation chain

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Field	Value
Generated file	cell3_to_29_locks_1783391141.pdf
Unix epoch	1783391141
Active derivation chain	galaxy-axis unit sphere -> native cell0 -> authoritative cell3 bridge -> cell3 Mass-Bridge -> reduced-mass checkpoint -> 29 witness rows
Primary cell3 inputs	A_ab, A_ac, and A_bc from authoritative cell3
Rendering discipline	ReportLab core fonts only; wrapped table cells; explicit table widths; repeatRows=1 on tables

Derivation provenance into the 29-row witness framework

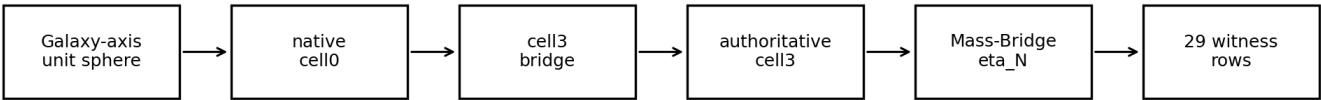


Figure 1. Derivation provenance into the 29-row witness framework.

Abstract

This paper carries the authoritative cell3 geometry into the 29-row native witness framework. The sequence begins with galaxy-axis data, produces a native cell0 scaffold, transitions to authoritative cell3, and then uses the primitive cell3 faces to derive the Mass-Bridge. From the Mass-Bridge, the paper derives the reduced-mass checkpoint, carrier-triangle quantities, native electromagnetic closure, the Gamma2 gravity scalar, the observer-rendered G row, and the downstream G-family rows.

The active cell3 face areas are $A_{ab} = 0.000754721789639889473$, $A_{ac} = 0.00101125575765694605$, and $A_{bc} = 0.00138578443174013771$. They produce $\eta_N = 1836.152673425999183$, $\alpha_N = 0.00729735256433142535$, $c_N = 137.035999177000008$, and $\mu_{ep} = 0.999455679424761656$. The same geometry gives $\Gamma_2 = 2.40058782142676733e-43$. The observer-context G row is rendered as $G = 6.674299999999994e-11 \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$; this row generates the gravitational couplings and Planck-family rows.

The paper is written as a reconstruction guide. Every table gives the formula used, the derived value, the comparison value, and the residual where a witness comparison is made.

1. Provenance and active inputs

The native galaxy derivation supplies cell0 as the galaxy-sourced scaffold. The cell0-to-cell3 bridge supplies the authoritative closure geometry. The Mass-Bridge is then computed from the authoritative cell3 primitive faces.

Dependency layer	Role in this paper
Native galaxy cell0	Establishes the galaxy-sourced rhombohedral scaffold.
Authoritative cell3 bridge	Provides the closure geometry used for all downstream computations.
Cell3 Mass-Bridge	Derives η_N from the scaled face quotient $1000 \cdot A_{bc}/A_{ab}$.
Current paper	Extends η_N into reduced mass, native witness rows, Γ_2 , G rows, and Planck rows.

2. Authoritative cell3 ledger

Quantity	Value	Role
a	0.020797058782	cell3 edge length
b	0.036494205131	cell3 edge length
c	0.0493085659	cell3 edge length
θ_{ab}	83.933483012313 deg	included angle for face ab
θ_{ac}	80.44813001833801 deg	included angle for face ac
θ_{bc}	50.363231813756 deg	included angle for face bc
A_{ab}	0.000754721789639889473	primitive face a x b
A_{ac}	0.00101125575765694605	primitive face a x c
A_{bc}	0.00138578443174013771	primitive face b x c

3. Carrier-triangle reconstruction

The three primitive faces define the carrier triangle. The mass channel, cross-leg channel, and fine/speed channel are reciprocal readings of this same face system.

$$\begin{aligned}
 C_m &= A_{bc}/A_{ab} \\
 \eta_N &= 1000 \cdot C_m \\
 R_{acab} &= A_{ac}/A_{ab} \\
 C_\alpha &= A_{bc}/A_{ac} \\
 \alpha_N &= A_{ac}/(100 \cdot A_{bc}) \\
 c_N &= 1/\alpha_N
 \end{aligned}$$

Quantity	Formula	Derived value
C_m	A_{bc}/A_{ab}	1.83615267342599919
η_N	$1000 \cdot C_m$	1836.152673425999183
R_{acab}	A_{ac}/A_{ab}	1.33990534199292166
C_α	A_{bc}/A_{ac}	1.37035999177000001

Quantity	Formula	Derived value
alpha_N	$A_{ac}/(100 \cdot A_{bc})$	0.00729735256433142535
c_N	$1/\alpha_N$	137.035999177000008
closure	$R_{acab} \cdot C_{\alpha} - C_m$	0.000000e-50

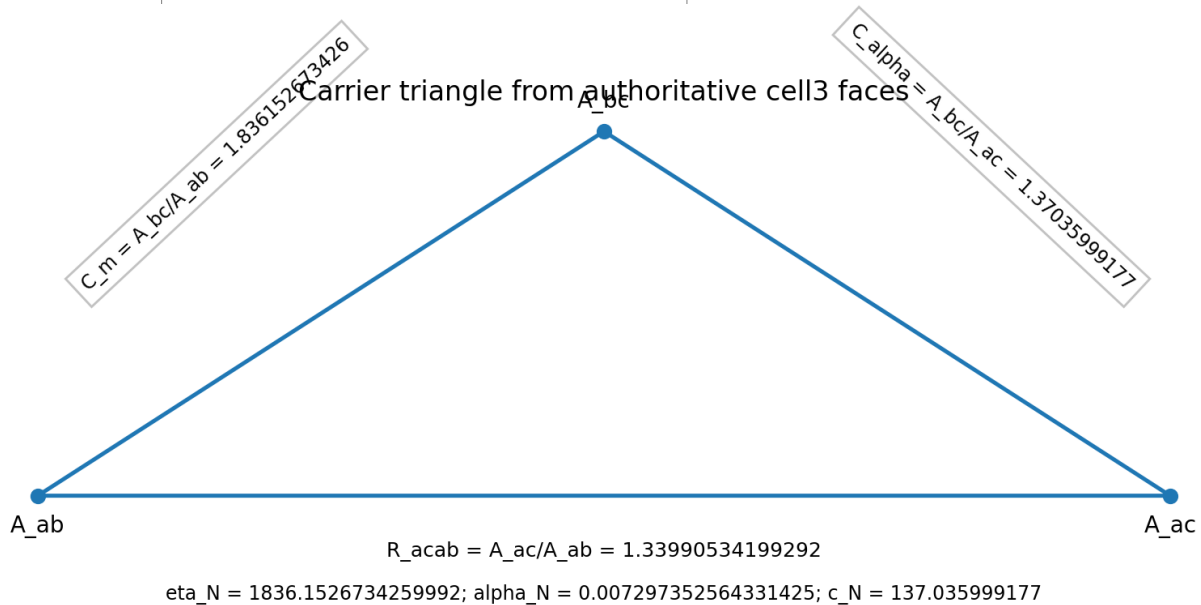


Figure 2. Carrier triangle computed from authoritative cell3 faces.

4. Reduced-mass checkpoint

The reduced-mass checkpoint converts the Mass-Bridge into the native two-body relative-motion factor. The inverse relation recovers the Mass-Bridge from the reduced-mass factor.

Object	Formula	Derived value
mu_ep	$\eta_N/(1+\eta_N)$	0.999455679424761656
eta recovery	$\mu_{ep}/(1-\mu_{ep})$	1836.15267342599918
S_mu	$1/(1+\eta_N)$	0.000544320575238397652

5. Non-G native witness rows

The following rows are generated from η_N , α_N , c_N , μ_{ep} , and their direct algebraic descendants. The comparison column is downstream of the derivation.

Row	CODATA witness name	Formula	Derived value	Witness value	Error ppt
A001	Reduced mass factor μ_{ep}	$\eta_N/(1+\eta_N)$	0.9994556794247617	0.999455679424762	-0.000398
A002	Proton/electron mass ratio η	η_N	1836.152673425999	1836.152673426	-0.000385
A003	Electron/proton mass ratio	$1/\eta_N$	0.000544617021488819	0.000544617021488819	0.000087
A006	Mass/fine denominator bridge R_{acab}	A_{ac}/A_{ab}	1.339905341992922	1.33990534199292	0.001301
A007	Fine-structure surrogate α_N	$A_{ac}/(100 \cdot A_{bc})$	0.007297352564331425	0.00729735256433142	0.000679
A008	Speed-of-light surrogate c_N	$1/\alpha_N$	137.035999177	137.035999177	0.000010
A011	Carrier/mass quotient C_m	A_{bc}/A_{ab}	1.836152673425999	1.836152673426	-0.000385
A012	Carrier/fine quotient C_α	A_{bc}/A_{ac}	1.37035999177	1.37035999177	0.000010
A014	Electron reduced Compton-like length	α_N	0.007297352564331425	0.00729735256433142	0.000679
A016	Electron full Compton-like length	$2 \cdot \pi \cdot \alpha_N$	0.04585061841351648	0.0458506184135165	-0.000351
A018	Classical-radius-like analogue	α_N^2	5.325135444815442e-05	5.325135444815440e-05	0.000443
A030	Thomson-cross-section-like analogue	$(8 \cdot \pi^2/3) \cdot \alpha_N^4$	2.375636132080968e-08	2.375636132080970e-08	-0.001022
A038	Rydberg-constant-like wavenumber	$\alpha_N/2$	0.003648676282165713	0.00364867628216571	0.000679
A039	Hydrogen Rydberg-like wavenumber	$\mu_{ep} \cdot \alpha_N/2$	0.003646690232592946	0.00364669023259295	-0.001278
A027	Magneton ratio	η_N	1836.152673425999	1836.152673426	-0.000385
A031	Vacuum-permeability-like $\mu_{0,N}$	$2 \cdot \alpha_N^2$	0.0001065027088963088	0.000106502708896309	-0.001434
A033	Vacuum-impedance-like $Z_{0,N}$	$2 \cdot \alpha_N$	0.01459470512866285	0.0145947051286628	0.003420

Representative non-G witness residuals

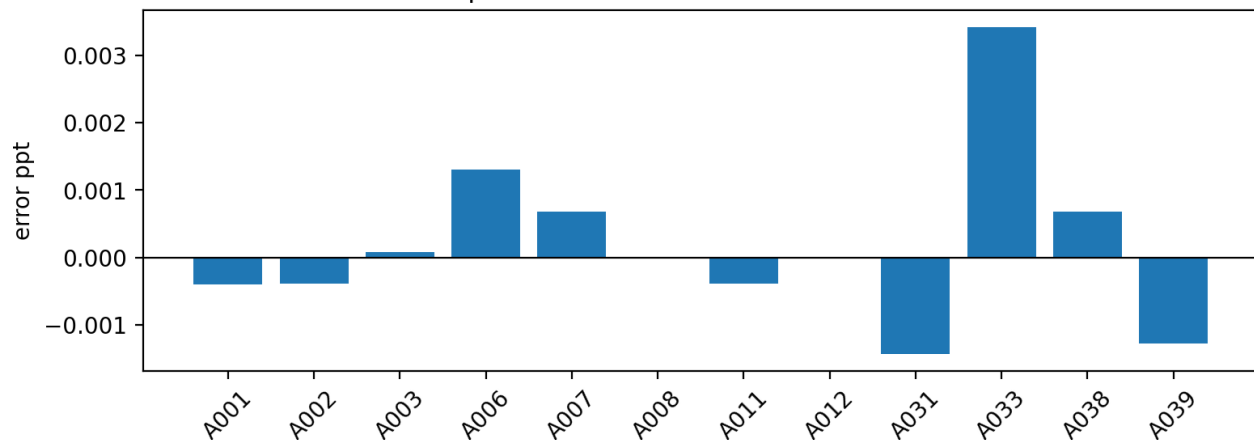


Figure 3. Representative non-G witness residuals in parts per trillion.

6. Native electromagnetic closure

The fine/speed channel also closes the native electromagnetic skeleton. With $\epsilon_{p0_N} = 1/2$ and $\mu_{0_N} = 2 \cdot \alpha_N^2$, the Maxwell-style speed relation returns c_N .

$$\begin{aligned}
 \mu_{0_N} &= 2 \cdot \alpha_N^2 \\
 \epsilon_{p0_N} &= 1/2 \\
 1/\sqrt{\mu_{0_N} \cdot \epsilon_{p0_N}} &= c_N \\
 Z_{0_N} \cdot c_N &= 2 \\
 R_{inf,N} \cdot c_N &= 1/2
 \end{aligned}$$

Identity	Computed value
$1/\sqrt{\mu_{0_N} \cdot \epsilon_{p0_N}}$	137.035999177000008
c_N	137.035999177000008
$Z_{0_N} \cdot c_N$	2
$R_{inf,N} \cdot c_N$	0.5

7. Gamma2 and observer-context G row

The gravity scalar is computed from the same authoritative cell3 faces. The source face form and the collapsed carrier-triangle form are algebraically equivalent.

$$\begin{aligned} \text{Gamma2} &= A_{ab}^9 A_{ac}^8 / (100^{11} A_{bc}^{11}) \\ \text{Gamma2} &= 10^3 A_{ab}^6 \alpha_N^8 \eta_N^{-3} \\ d \ln(\text{Gamma2}) &= 6 d \ln(A_{ab}) + 8 d \ln(\alpha_N) - 3 d \ln(\eta_N) \end{aligned}$$

Expression	Value
source face formula	2.40058782142676733e-43
collapsed formula	2.40058782142676733e-43
fingerprint	(6, 8, -3)

The G-family rows use the baseline observer-context rendering packet. The rendered scalar G row is the entry point for gravitational coupling and Planck-family rows.

Object	Value
Frame	SGA-2020 / ICRS / J2000
Epoch	2026-06-17T15:06:11Z
Reference direction v_hat	(-0.97077086, 0.20734801, -0.12087493)
f_obs	1.447739
beta_obs^2	1.555741578671e-06
G row	6.674299999999994e-11 m^3 kg^-1 s^-2

G-family rendering sequence used by the witness framework

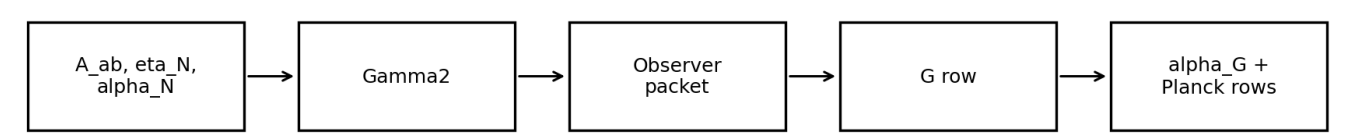


Figure 4. Observer-context G-family rendering sequence.

8. G-family witness rows

The G-family rows are computed from the observer-rendered G row, SI action and light-speed renderings, and the mass bridge for proton/electron grading.

Row	CODATA witness name	Formula	Derived/rendered value	Witness value	Error ppt
A048	Newtonian gravitational constant G	G	6.674299999999999e-11 m ³ kg ⁻¹ s ⁻²	6.674299999999999e-11 m ³ kg ⁻¹ s ⁻²	0.000000
A050	Electron gravitational coupling alpha_G,e	$G*m_e^2/(\hbar*c)$	1.751809398790771e-45	1.751809398790770e-45	0.000619
A051	Electron-proton gravitational coupling alpha_G,ep	$G*m_e*m_p/(\hbar*c)$	3.216589510922467e-42	3.216589510914630e-42	2.436388
A052	Proton gravitational coupling alpha_G,p	$G*m_p^2/(\hbar*c)$	5.906149429794315e-39	5.906149429765540e-39	4.872054
A053	Planck length l_P	$\sqrt{\hbar*c/G}$	1.616255024423705e-35 m	1.616255024423710e-35 m	-0.002878
A054	Planck time t_P	$\sqrt{\hbar*c/G}$	5.391246448313605e-44 s	5.391246448313610e-44 s	-0.001082
A055	Planck mass m_P	$\sqrt{\hbar*c/G}$	2.176434342717898e-08 kg	2.176434342717900e-08 kg	-0.000783
A056	Planck/proton mass ratio	m_P/m_p	1.301211175667475e+19	1.301211175670650e+19	-2.439610
A057	Planck energy E_P	m_P*c ²	1.956081636698373e+09 J	1.956081636698370e+09 J	0.001484
A058	Planck temperature T_P	E_P/k_B	1.416784162157343e+32 K	1.416784162157340e+32 K	0.001835
A059	Planck force F_P	c ⁴ /G	1.210255564338206e+44 N	1.210255564338210e+44 N	-0.003067
A060	Planck power P_P	c ⁵ /G	3.628254904411280e+52 W	3.628254904411280e+52 W	0.000018

9. Complete 29-row witness summary

This table collects the full witness set used in the paper. Rows A001-A033 are non-G native rows. Rows A048-A060 are observer-context G-family rows.

Row	CODATA witness name	Derived/rendered value	Witness value	Error ppt
A001	Reduced mass factor mu_ep	0.999455679424762	0.999455679424762	-0.000398
A002	Proton/electron mass ratio eta	1836.152673426	1836.152673426	-0.000385
A003	Electron/proton mass ratio	0.000544617021488819	0.000544617021488819	0.000087
A006	Mass/fine denominator bridge R_acab	1.33990534199292	1.33990534199292	0.001301
A007	Fine-structure surrogate alpha_N	0.00729735256433143	0.00729735256433142	0.000679
A008	Speed-of-light surrogate c_N	137.035999177	137.035999177	0.000010
A011	Carrier/mass quotient C_m	1.836152673426	1.836152673426	-0.000385
A012	Carrier/fine quotient C_alpha	1.37035999177	1.37035999177	0.000010
A014	Electron reduced Compton-like length	0.00729735256433143	0.00729735256433142	0.000679
A016	Electron full Compton-like length	0.0458506184135165	0.0458506184135165	-0.000351
A018	Classical-radius-like analogue	5.32513544481544e-05	5.32513544481544e-05	0.000443
A030	Thomson-cross-section-like analogue	2.37563613208097e-08	2.37563613208097e-08	-0.001022
A038	Rydberg-constant-like wavenumber	0.00364867628216571	0.00364867628216571	0.000679
A039	Hydrogen Rydberg-like wavenumber	0.00364669023259295	0.00364669023259295	-0.001278
A027	Magneton ratio	1836.152673426	1836.152673426	-0.000385
A031	Vacuum-permeability-like mu0,N	0.000106502708896309	0.000106502708896309	-0.001434
A033	Vacuum-impedance-like Z0,N	0.0145947051286629	0.0145947051286628	0.003420
A048	Newtonian gravitational constant G	6.674300000000000e-11 m ³ kg ⁻¹ s ⁻²	6.674300000000000e-11 m ³ kg ⁻¹ s ⁻²	0.000000
A050	Electron gravitational coupling alpha_G,e	1.75180939879077e-45	1.75180939879077e-45	0.000619
A051	Electron-proton gravitational coupling alpha_G,ep	3.21658951092247e-42	3.21658951091463e-42	2.436388
A052	Proton gravitational coupling alpha_G,p	5.90614942979432e-39	5.90614942976554e-39	4.872054
A053	Planck length l_P	1.61625502442371e-35 m	1.61625502442371e-35 m	-0.002878
A054	Planck time t_P	5.39124644831360e-44 s	5.39124644831361e-44 s	-0.001082

Row	CODATA witness name	Derived/rendered value	Witness value	Error ppt
A055	Planck mass m_P	2.17643434271790e-08 kg	2.17643434271790e-08 kg	-0.000783
A056	Planck/proton mass ratio	1.30121117566748e+19	1.30121117567065e+19	-2.439610
A057	Planck energy E_P	1.95608163669837e+09 J	1.95608163669837e+09 J	0.001484
A058	Planck temperature T_P	1.41678416215734e+32 K	1.41678416215734e+32 K	0.001835
A059	Planck force F_P	1.21025556433821e+44 N	1.21025556433821e+44 N	-0.003067
A060	Planck power P_P	3.62825490441128e+52 W	3.62825490441128e+52 W	0.000018

10. Student reconstruction ledger

Object	Formula	Value
A_ab	a x b	0.000754721789639889473
A_ac	a x c	0.00101125575765694605
A_bc	b x c	0.00138578443174013771
eta_N	1000*A_bc/A_ab	1836.152673425999183
mu_ep	eta_N/(1+eta_N)	0.999455679424761656
alpha_N	A_ac/(100*A_bc)	0.00729735256433142535
c_N	1/alpha_N	137.035999177000008
Gamma2	10^3*A_ab^6*alpha_N^8*eta_N^-3	2.40058782142676733e-43
G	observer-context G row	6.674299999999994e-11 m^3 kg^-1 s^-2

Conclusion

The derivation chain is now continuous: galaxy-axis data produce native cell0; the cell0-to-cell3 bridge supplies authoritative cell3; authoritative cell3 supplies the primitive faces; the primitive faces supply the Mass-Bridge and carrier triangle; the Mass-Bridge supplies reduced mass; and the same native quantities generate the 29-row witness framework.